MIL-H-14516E
30 December 1987
SUPERSEDING
MIL-H-14516D
17 September 1970

MILITARY SPECIFICATION

HEATER, SPACE: OIL-FIRED

70,000 BTU PER HOUR,

THEATER OF OPERATIONS TYPE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers one size of oil-fired, general-purpose space heater having a heat output of 70,000 Btu/hr at 0.05-inch water gage draft. The space heater utilizes a standard military rectangular 5-gallon fuel can as a fuel tank.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Documents. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development and Engineering Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4520

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SPECIFICATIONS

FEDERAL

FF-N-105	-	Nails, Brads, Staples and Spikes: Wire, Cut and Wrought.
QQ-S-781	-	Strapping, Steel, and Seals.
QQ-W-461	-	Wire, Steel, Carbon (Round, Bare and Coated).
UU-T-81	-	Tags, Shipping and Stock.
VV-F-800	-	Fuel Oil, Diesel.
PPP-B-591	-	Boxes, Shipping, Fiberboard, Wood-Cleated.
PPP-B-601	-	Boxes, Wood, Cleated-Plywood.
PPP-B-636	-	Boxes, Shipping, Fiberboard.
PPP-C-843	-	Cushioning Material, Cellulosic.
PPP-F-320	-	Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.
PPP-T-60	-	Tape, Packaging, Waterproof.
PPP-T-97	-	Tape, Pressure-Sensitive Adhesive, Filament Reinforced.

MILITARY

MIL-P-116	-	Preservation, Methods of.
MIL-L-21260	-	Lubricating Oil, Internal Combustion Engine, Preservative and Break-In.

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.
MIL-STD-130	-	Identification Marking of US Military Property.
MIL-STD-248	-	Welding and Brazing Procedure and Performance Qualification.
MS24629	-	Screw, Tapping-Thread Cutting, Types D, F, G or T Pan Head, Cross Recessed, Steel, Carbon, Cadmium Plated.
MS27183	-	Washer, Flat-Round, Steel, Cadmium Plated, General Purpose.
MS35206	-	Screw, Machine-Pan Head, Cross-Recessed, Carbon Steel, Cadmium Plated, UNC-2A.
MS35338	-	Washer, Lock-Spring, Helical, Regular (Medium) Series.

DRAWINGS

U.S. ARMY NATICK RESEARCH, DEVELOPMENT, AND ENGINEERING CENTER

5-13-2851	-	Heater, Space, Oil Fired, 70,000 Btu/hr, Drawing and Part List

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(Copies of documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity).

2.2 Other publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issues of the nongovernment documents which are current on the date of the solicitation.

UNDERWRITERS LABORATORIES, INC.

UL 507 - Electric Fans
UL 896 - Oil-Burning Stoves

(Application for copies should be addressed to the Underwriters Laboratories, Inc., 333 Pfingsten Road, Northbrook, IL 60062.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS

Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

ASHRAE - AMCA STANDARD

Ashrae 51-1985/AMCA 210-85-Laboratory Methods of Testing Fans for Ratings.

(Applications for copies should be addressed to Air Moving and Conditioning, Inc., 30 West University Drive, Arlington Heights, IL 60004, or American Society of Heating, Refrigerating and Air-Conditioning Engineers, 1791 Tullie Circle, N.E. Atlanta, GA 30329.)

AMERICAN WELDING SOCIETY

AWS Standard B3.0, Welding Procedure and Performance Qualification.

(Application for copies should be addressed to the American Welding Society, 550 NW Lejeune Road, P.O. Box 351040, Miami, FL 33135.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

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2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3, 6.2, and 6.3).

3.2 Codes and standards. The space heater shall conform to the applicable requirements of UL 507 and UL 896. Satisfactory evidence that these requirements have been met shall be submitted to the contracting officer or his authorized representative prior to the start of production (see 4.4.6).

3.3 Materials and components. Materials and components shall be as specified herein and as shown on the applicable drawings. Materials and components not definitely specified shall be of the quality normally used by the contractor in the manufacture of his space heater, provided the completed item complies with all provisions of this document (see 6.4).

3.3.1 Fan. The construction of the fan shall be in accordance with UL 507 (see 4.4.1.2).

3.4 Design and construction. The heater shall be in accordance with drawings listed on Drawing 5-13-2851 and as specified herein. The drawings forming a part of this document are engineering design drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Unless otherwise specified in the contract or purchase order, any data (e.g., shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the supplier or obtained from a vendor to support fabrication and manufacture of the production item shall be made available upon request for inspection by the contracting officer or his designated representative (see 6.2). When specified (see 6.2), a smoke jack assembly and a fan assembly shall be furnished. The heater shall have a minimum operational life of 3,000 hours before requiring repairs or replacement of parts, shall operate for a minimum of 300 hours prior to cleaning the burner pot, and shall operate for 600 hours without requiring maintenance other than cleaning of the burner pot each 300 hours.

3.4.1 Heater output. The heaters shall develop the heat outputs specified in table I, without visible smoke, when burning grade DF-2 Continental US diesel fuel conforming to VV-F-800. Output and performance shall also conform to UL 896 except for fuel flow and control valve calibration.

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TABLE I. Output

Btu/hr	Fuel rate (cubic centimeters/minute at 4.00 centistokes at 70°F)	Draft (inches of water)
70,000 (minimum high-fire)	48 ± 2	0.05
20,000 (maximum low-fire)	6 ± 1	0.05

3.4.2 Seepage and burn-off. The heater shall conform to the seepage and burn-off requirements specified in UL Subject 896 (see 4.5.4).

3.4.3 Tilted operation. A tilt of 2 degrees in any direction from horizontal shall not affect starting or specified operation of the heater. (see 4.5.5).

3.4.4 Fuel control valve flow rate. The flow rate of the fuel control valve component shall be as specified on Drawing No. 13208E6211 (see 4.4.1.1).

3.4.5 Siphon pump and fuel control valve assembly flow test. When tested in accordance with 4.5.2, the fuel shall flow freely when the valve is set at each of the "open" position settings and shall stop flowing when the valve is in the off position.

3.4.6 Smoke reading. When operated at the high-fire fuel rate, the smoke reading shall be less than the No. 7 spot on the Shell-Bacharach Scale (see 4.5.7).

3.4.7 Combustion efficiency. When operated at the high-fire fuel rate and at an acceptable smoke reading, the combustion efficiency shall be greater than 70 percent. The efficiency shall be determined by the carbon dioxide (CO_2) and the temperature increase of the flue gas (see 4.5.8).

3.5 Marking for identification. All parts shall be marked for identification in accordance with MIL-STD-130.

3.6 Data-name plates. The space heater shall be furnished with a data-name plate in accordance with the applicable requirements of MIL-STD-130 except the requirements for: (a) Methods of applying, (b) Identification tags, (c) Information not required and (d) Optional marking information shall not apply. The data plates shall be made of minimum 20 gauge corrosion-resisting metal and attached to the space heater by rivets, screws or welding in such a manner as to meet the applicable National Sanitation Foundation sanitary requirements for this equipment. The plate shall contain the following information which shall be stamped, engraved or applied by photosensitive means:

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National stock number
Procurement instrument identification number
Specification data
Manufacturer's name, address and telephone number
Supplier's name, address and telephone number (List only if different from manufacturer)
Manufacturer's model number
Government approved manual number (See 6.2)

Each plate shall be placed so that it is readily visible to the operator during normal operating use and so as to not adversely affect the life and utility of the space heater.

3.7 Workmanship.

3.7.1 Metal fabrication. Sheet metal and structural members used in fabrication shall be straight and free from laminations. Sheet metal and structural members may be cut by shearing or sawing. All burrs and rough edges shall be removed. Corners shall be square and true. All bends shall be made to a radius which will not cause cracking at the extreme fibers of the piece being bent, and all bends except for minor details shall be made by use of metal dies or fixtures to insure uniformity in size and shape.

3.7.2 Bolted connections. Bolt holes shall be formed or drilled and shall have the burrs removed. Washers and lockwashers shall be provided, and all bolts, nuts, and screws shall be tightened so that they will not work loose under vibration.

3.7.3 Riveted connections. Rivet holes shall be punched or drilled and shall have the burrs removed. Rivet heads shall be full, concentric with the rivet holes, and in full contact with the surface of the members.

3.7.4 Welders and welding.

3.7.4.1 Welders. All welders shall maintain current qualifications appropriate to their job under one of the following qualification standards:

(a) Welding Procedure and Performance Qualification of the American Welding Society AWS B3.0
(b) ASME Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.
(c) MIL-STD-248.

3.7.4.2 Welding. The surfaces of parts to be welded shall be free from scale, paint, grease, and other foreign matter. Welding shall be in accordance with AWS B3.0.

3.7.5 Jigs and fixtures. Shop-fabricated components shall be assembled in steel jigs or frames and joined while held in position. Jigs or frames shall minimize distortion.

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3.7.6 Interchangeability. All like components shall be interchangeable with respect to installation, performance, and quality.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this document shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the document shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Responsibility for dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for assuring that all specified dimensions have been met. When dimensions cannot be examined on the end item, inspection shall be made at any point, or at all points in the manufacturing process necessary to assure compliance with all dimensional requirements (see 4.4.4).

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 6.2), it shall be examined for the defects in table II and 4.4.4, and tested as specified in 4.4.5. The presence of any defect or failure of any test shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

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4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.4.1.1 Parts and components. Parts and components detailed on drawings shall be inspected in accordance with the Supplementary Quality Assurance Provisions (SQAP) shown on the drawings (see 6.6). The drawings specify the characteristics requiring SQAP inspection, the sampling plan, and the basis for acceptance and rejection. A certificate of compliance indicating the fuel control valve meets the requirements of 3.4.4 will be acceptable.

4.4.1.2 Fan. The fans shall be inspected to determine compliance with UL-507 and tested in accordance with ASHRAE 51-1985/AMCA 210-85 to determine fan capacity for 0 pressure. Nonconformance with the 650 CFM's specified on the drawings shall constitute failure of this test (see 3.3.1).

4.4.1.3 Testing of constant level valve. The constant level valve shall be tested in accordance with 4.5.2. Failure to comply with the siphon pump and fuel control valve assembly requirements of 3.4.5 shall be cause for rejection of the lot.

4.4.2 In-process inspection. Inspection shall be made at any point or during any phase of the manufacturing operation to determine whether construction details which cannot be examined in the finished product are in accordance with specified requirements. Whenever nonconformance is noted, corrections shall be made to the items affected and lot in-process. Items which cannot be corrected shall be removed from production.

4.4.3 End item visual examination. The end item shall be examined for the defects listed in table II. The lot size shall be expressed in units of heaters. The sample unit shall be one heater. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5 for major and 6.5 for total (major and minor combined) defects.

TABLE II. End item visual defects

Examine	Defect	Classification Major	Minor
Finish	Color not as specified	X	
	Does not meet NSF standards	X	
	Not smoothly applied, blisters flaking, peeling or scratches		X
Construction and workmanship	Component missing not in proper location, or not as specified	X	
	Fabrication not as specified	X	

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TABLE II. End item visual defects (cont'd)

Examine	Defect	Classification Major	Classification Minor
Construction and workmanship (cont'd)	Bolted and riveted connections not as specified	X	
	Washers or lockwashers missing		X
	Welding not as specified	X	
Identification marking	Missing, incomplete or not as specified	X	
	Illegible, incomplete, incorrect or wrong location		X
Data name plates	Omitted or not as specified		X
	Information incomplete or not legible		X
	Not located so as to be visible to operator		X

4.4.4 End item dimensional examination. Examination shall be made of the heaters for the dimensions specified with an asterisk on the Drawings. Any dimension not within specified requirements shall be classified as a defect. The lot size shall be expressed in units of heaters. The sample unit shall be one completely fabricated heater. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

4.4.5 End item testing. The end item shall be tested for the characteristics listed below. The lot size shall be expressed in units of space heaters. The sample unit shall be one heater. The service life test shall be the last test performed. The inspection level shall be S-1. Failure of any test shall be cause for rejection of the lot.

Characteristic	Requirement Paragraph	Test Paragraph
Combustion efficiency	3.4.7	4.5.3 and 4.5.8
Smoke	3.4.6	4.5.7
Heater output	3.4.1	4.5.9
Seepage and burn-off	3.4.2	4.5.4
Tilted operation	3.4.3	4.5.5
Service life	3.4	4.5.6

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4.4.6 Codes and standards examination. Certifications, certified test reports, or listing marks for codes and standards, as applicable, submitted in accordance with 3.2, shall be examined and validated as proof of compliance.

4.4.6.1 Underwriters' Laboratories, Inc. (UL). Acceptable evidence of meeting the applicable requirements of the Underwriters' Laboratories standards specified in 3.2 shall be the UL label, a UL listing mark, or a certified test report from a recognized independent laboratory, acceptable to the Government, indicating that the space heater conforms to the applicable requirements of the specified UL 507 and UL 896 (see 3.2).

4.4.7 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

Examine	Defect
Marking (including match marking where required)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application
Materials	Any component missing, damaged, or not as specified
	Preservatives (where required) missing or not as specified 1/
Workmanship	Inadequate application of components, such as: loose strapping, inadequate stapling, improper taping, or incomplete sealing or closure of flap
	Bulged or distorted container

1/ Contents of one shipping container from the sample for each lot shall be examined for this condition.

4.5 Methods of inspection.

4.5.1 Conditions. Unless otherwise specified (see 6.2), tests on the heater shall be conducted utilizing all components except the auxiliary fan. All operating tests on the heater shall be conducted while burning Continental US diesel fuel conforming to VV-F-800, grade DF-2. No braces, tiedown devices, or other items that are not inherently a part of the operating heater shall be employed. No alignment of components, replacement of components, or other servicing shall be performed on the first article or production heaters after testing of the heater has been initiated. Tests on the auxiliary fan shall be made without the heater. Efficiency tests shall be conducted with the heater, stove pipes, elbow, and tee installed as shown on the applicable drawing, and

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tested in accordance with installation test of UL 896. The heater shall be in operation at the prescribed fuel rate with the draft regulator adjusted to 0.05 inches of water column for a period not less than 15 minutes to allow for the heater to stabilize. Three readings shall be taken at intervals of 15 minutes. The average reading shall be taken for the final determination.

4.5.2 Fuel valve flow rate. Prior to assembly on the heater, the constant level valve, with the strainer in place and assembled with the siphon tube assembly and the pot burner fuel line, shall be tested for flow rate. The valve control knob shall be turned to the maximum flow setting and the fuel allowed to flow to purge air from the system. After purging, the fuel shall flow freely through the valve at each setting on the valve. The flow of fuel shall stop when the valve control knob is turned to the "off" position. Inability of the fuel to flow freely when the valve is set at the open positions or to stop flowing when the valve is turned to the "off" position shall constitute a test failure.

4.5.3 Efficiency. Test the heater for efficiency and compute the output as follows:

a. Assemble the heater per Drawing 5-13-2851. Use eight lengths of stove pipe and the rain cap. The assembly shall be level.

b. Use the constant level valve with strainer, previously tested in accordance with 4.5.2.

c. Use the siphon tube assembly to start the flow of fuel to the constant level valve. Adjust the flow to the maximum setting.

d. Determine the efficiency and output in accordance with UL 896 and 4.5.8.

e. Determine if the flow of fuel ceases, when the valve is shut off, and when the pot is flooded.

f. Readings and calculations shall be reported on the Data and Report Sheet (see table III).

Inability to start or to operate as required on medium flow, nonconformance to 3.4.7, evidence of oil leaks, or inability to shut off shall constitute failure of this test.

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4.5.4 Seepage and burn-off. Test the heater in accordance with UL Subject 896, test No. 11. Nonconformance to 3.4.2 shall constitute failure of this test.

4.5.5 Tilted operation. Tilt the heater in four different positions, 90 degrees apart, and at a slope of 2 degrees from the horizontal. Start and operate in each position for a period of 30 minutes. Inability of the heater to start, operate, or restart shall constitute failure of this test (see 3.4.3).

4.5.6 Service life. With the fan (when furnished) installed not more than 3 feet above the heater, operate the heater and fan for 600 hours actual operational time on a cycling schedule of 16 hours at high fire followed by 8 hours at low fire (20,000 or less Btu/hr output). No maintenance except cleaning of the burner pot each 300 hours is permitted. A military rectangular 5-gallon fuel can and a siphon and pump shall be used throughout this test. Successful completion of this test shall constitute compliance with the 3,000-hour minimum operating life requirement (see 3.4).

4.5.6.1 Failure criteria. One or more of the following shall constitute failure of this test:

a. Inability of the heater to complete 300 hours before requiring cleaning of the burner pot.
b. Inability of the heater to complete 600 hours without maintenance other than cleaning of the burner pot.
c. Incorrect function of the constant-level valve, burn through or cracking or breaking of metal, warping affecting operation, or a variation of more than 15 percent in per-hour fuel consumption.
d. Incorrect function of fan components.
e. Dripping of oil from motor sealed bearings.
f. Any operating difficulty, malfunction, or permanent deformation affecting operation or breakage.

4.5.7 Smoke reading. Five smoke readings shall be taken by the smoke measuring instrument specified (see 6.7).

4.5.8 Combustion efficiency.

4.5.8.1 Carbon dioxide. The carbon dioxide samples shall be taken in accordance with the installation test of UL 896, using a CO_2 analyzer.

4.5.8.2 Temperature readings. The ambient and the flue gas temperature shall be taken in accordance with the installation test of UL 896.

4.5.8.3 Combustion efficiency. The following formula shall be used for determining the combustion efficiency:

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$$E = 1 - \left[\frac{3171 \div 2.69\ Tf \div \left[\frac{216.2 \div 0.77}{CO_2}\right](Tf - Ta) = 5.38\ Ta}{HHV} \right]$$

E = Combustion efficiency (percent)
Tf = Flue gas temperature (°C)
Ta = Ambient temperature (°C)
CO_2 = Carbon dioxide (percent)
HHV = High heat value of the fuel (Joules per gram)

4.5.9 Heater output. The fuel as specified in 3.4.1 shall be assumed to have a gross heating value of 19750 Btu's per pound. The combustion efficiency (see 3.4.7) as determined by 4.5.8.3 shall be used to determine the output as follows:

Heater output (Btu/hr) = Efficiency x Input (Btu/hr)

5. PACKAGING

5.1 Preservation. Preservation shall be level A, B or C as specified (see 6.2).

5.1.1 Disassembly and matchmarking. The heater components shall be disassembled as required to accomplish the unit packing specified in 5.1.1.1.8. Disassembled components shall be matchmarked to facilitate reassembly. Matchmarking information shall be on cloth tags conforming to UU-T-81, type A, attached to mating parts.

5.1.1.1 Level A.

5.1.1.1.1 Preservatives. Cleaning, application and type of preservation, and method of preservation shall be in accordance with MIL-P-116 as specified herein.

5.1.1.1.2 Unprotected surfaces. Unpainted and untreated metal surfaces of the heater and components requiring the application of a contact preservative in accordance with MIL-P-116 shall be coated with type P-1 preservative.

5.1.1.1.3 Fuel system and burner. Interior surfaces of the siphon tube, the valve, and the fuel lines shall be coated with preservative conforming to MIL-L-21260, type I, grade 30. Excess preservative shall be drained.

5.1.1.1.4 Auxiliary fan assembly. The box containing the fan assembly shall be waterproof sealed with tape in accordance with the appendix to PPP-B-636.

5.1.1.1.5 Leveling screws, pins, straps, and other fasteners. Screws, pins, straps, and fasteners shall be coated with type P-7 preservative and preserved in accordance with MIL-P-116, method IC-1.

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5.1.1.1.6 Instructional material. Instructional material for each heater shall be preserved together in accordance with MIL-P-116, method IC-1.

5.1.1.1.7 Technical publications. When technical publications are furnished, they shall be individually preserved in accordance with MIL-P-116, method IC-1.

5.1.1.1.8 Unit packing. Heaters without fan and smoke jack assembly shall be packed in accordance with table IV and figures 1 and 2, level A. Heaters with fan and smoke jack assembly shall be packed in accordance with table V and figures 3 and 4, level A.

5.1.1.2 Level B. Preservation shall be as specified for level A except that unit packing for heaters without fan and smoke jack assembly shall be in accordance with table IV and figures 1 and 2, level B. Unit packing for heaters with fan and smoke jack assembly shall be in accordance with table V and figures 3 and 4, level B.

5.1.1.3 Level C. The heater components, technical publications and instructional material shall be preserved to afford protection against deterioration and damage from the supplier to the initial destination. Unit packing for heaters without fan and smoke jack assembly shall be in accordance with table IV and figures 1 and 2, level C. Unit packing for heaters with fan and smoke jack assembly shall be in accordance with table V and figures 3 and 4, level C.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A packing. Heaters without fan and smoke jack assembly shall be packed in accordance with table IV and figures 1 and 2, level A. Heaters with fan and smoke jack assembly shall be packed in accordance with table V and figures 3 and 4, level A. When technical publications are furnished, they shall be packed with the respective item.

5.2.2 Level B packing. Heaters without fan and smoke jack assembly shall be packed in accordance with table IV and figures 1 and 2, level B. Heaters with fan and smoke jack assembly shall be packed in accordance with table V and figures 3 and 4, level B. When technical publications are furnished, they shall be packed with the respective item.

5.2.3 Level C packing. Heaters without fan and smoke jack assembly shall be packed in accordance with table IV and figures 1 and 2, level C. Heaters with fan and smoke jack assembly shall be packed in accordance with table V and figures 3 and 4, level C. When technical publications are furnished, they shall be packed with the respective item.

5.3 Marking. In addition to any special marking required by the contract or purchase order, shipping containers shall be marked in accordance with MIL-STD-129.

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6. NOTES

6.1. Intended use. The heaters are intended for use primarily for heating prefabricated and temporary-type buildings.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this document.
b. When a first article is required (see 3.1, 4.3, and 6.3).
c. When smoke jack assembly is required (see 3.4).
d. When fan assembly is required (see 3.4).
e. Government approved manual number (see 3.6).
f. Conditions of test, if different (see 4.5.1).
g. Level of preservation and packing required (see 5.1 and 5.2). Level B preservation and packing is intended to provide limited but economical protection and should be specified only when it is determined the heaters will be held in covered storage.
h. Required data will be made available unless otherwise specified (see 3.4).

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all acquisition instruments regarding arrangements for selection, inspection, and approval of the first article.

6.4 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the document (see 3.3).

6.5 Supplementary quality assurance provisions (SQAP). The contracting officer should require the contractor to maintain records of all SQAP inspections. A suggested paragraph is as follows:

"The contractor shall maintain complete records of all examinations and tests performed to verify the requirements of classified SQAP characteristics. The records shall include as a minimum, lot size, sample size, drawing requirement, actual measurement, number and type of deficiencies found, quantity approved, quantity rejected, corrected action taken when applicable."

6.6 Definitions.

6.6.1 Supplementary quality assurance provisions (SQAP). A SQAP is a contractual requirement that supplements section 4 of the document. SQAP indicates the minimum requirements which must be inspected on the product drawings to verify the design objectives of the product and assure interchangeability of repair parts.

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6.7 Smoke and carbon dioxide measuring instruments. Smoke and carbon dioxide measuring instruments manufactured by Bacharach Instrument Div., American Basch Arma Corp., have been found to be satisfactory.

6.8 Subject term (key word) listing.

Fuel Burning Space Heater
Oil Fired Air Heater
Oil Fired Unit-Heater
Radiant Space Heater
Space Heater

6.9 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:

Army - GL
Navy - YD
Air Force - 99

Preparing activity:

Army - GL

Project No. 4520-0307

Review activities:

Army - ME
DLA - CS
Air Force - 84

User activity:

Navy - MC

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TABLE III. OIL-HEATER RATING TEST
Data and Report Sheet

Manufacturer's test No. --------

Oil heater ----------
(Make) (Model, type, or no.) (Number and size of burners)
Fuel used for test ----------
(Brand) (API gravity No. at 60°F)
Date of test ---------- Tested by ----------

Test data	Three sets of readings at equal intervals not less than 20 minutes apart.			Average
1. Draft ---------- (in. of water) --	----	----	----	-----
2. Room temperature ---------- °F --	----	----	----	-----
3. Smoke-meter readings --------- (spot #_)	----	----	----	-----
4. Fuel-temperature readings ------ (°F) --	----	----	----	-----
5. Time intervals used for fuel readings --	----	----	----	-----
6. Fuel input readings (specify units used)	----	----	----	-----
7. Fuel-oil input rate ---------- (lb/hr avg) --				-----
8. Gross heating value of oil ----------(Btu/lb) --				-----
9. Gross heat input ----------(Btu/hr) --				-----
10. Flue-gas temperature ----------(°F) ---	----	----	----	-----
11. Flue-gas temp. rise above room temp(°F)	----	----	----	-----
12. CO_2 in dry flue gas ----------(%)-------	----	----	----	-----
13. CO ---------- (%) -----				-----
14. Efficiency for complete combustion in accordance with the efficiency formula in 4.5.8.3 (%) -----				-----
15. Heat output in accordance with the heater output formula in 4.5.9 (Btu/hr) --				-----
16. Fuel-input rate for low-fire smoke test ---------- (lb/hr) --				-----
17. Smoke-meter readings at 20-min intervals (low-fire test) ---------- (spot #-)	----	----	----	-----

We hereby certify that the above is a true and accurate copy of the data obtained when the above heater was tested in accordance with this document.

For _____
Address _____
Date _____ By _____

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TABLE IV. Preservation and packing material for heaters without fan and smoke jack assembly

FIND NO.	IDENTIFICATION	DESCRIPTION	LEVEL A	LEVEL B	LEVEL C
1	PPP-B-601, overseas type, style A, type 3 load (see Note 1)	Box, Plywood, Cleated	1		
1	PPP-B-591, class II, style A (see note 1)	Box, Fiberboard, Cleated		1	
2	PPP-B-636, V3c, style RSC	Box, Fiberboard	1	1	
2	PPP-B-636, type CF, class domestic, variety SW, grade 275, style RSC	Box, Fiberboard			1
3	PPP-F-320, type CF, class domestic, variety SW, grade 125	Fiberboard	1	1	1
4	Base, side members	Lumber, softwood, 1x6 nominal	2	2	2
5	Base, cross members	Lumber, softwood, 1x3 nominal	2	2	2
6	FF-N-105, type II, style 4, table Vb	Nail, 6d	AR	AR	AR
7	MS27183-11, .312 IDx.734 ODx.065 THK	Washer, flat-round, steel, cad plated, general purpose	4	4	4
8	MS35338-45, 5/16 nominal size	Washer, lockspring, helical, regular series, stl, cad plated	4	4	4
9	MS35206-301, 5/16-18UNC-2Ax1-3/4 long	Screw, mach, pan hd, cross recessed, cs, cad plated	4	4	4
10	QQ-W-461, finish 5, soft, class 1, .080 DIA	Wire, light, soft tensile, light zinc coated	AR	AR	AR
11	PPP-F-320, Type CF, class domestic, variety SW, grade 125	Fiberboard	AR	AR	AR
12	PPP-F-320, Type CF, class domestic, variety TW, grade 1100	Fiberboard	1	1	1
13	PPP-F-320, Type CF, class domestic, variety TW, grade 1100	Fiberboard	1	1	1
14	QQ-S-781, class I, type I or IV, Finish A	Steel strapping, flat, .035 x 3/4 W	AR	AR	AR

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TABLE IV. Preservation and packing material for heaters without fan and smoke jack assembly (cont'd)

FIND NO.	IDENTIFICATION	DESCRIPTION	LEVEL A	LEVEL B	LEVEL C
15	PPP-T-60, type IV, class I	Tape, pressure sensitive	AR	AR	AR
16	Blocking	Lumber, softwood, 1x3 nominal	2	2	2
17	QQ-S-781, type I, class A	Steel strapping, flat, .023 x 1/2 W	AR	AR	AR
18	Blocking	Lumber, softwood, 2x2 nominal x 3LG	2	2	2
19	PPP-C-843, type II, class A	Cushioning	AR	AR	AR
20	PPP-T-97, type III	Tape, pressure sensitive, 3/4 W	AR	AR	AR

NOTES:
1. Top to be inverted.
2. AR denotes "as required".

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TABLE V. Preservation and packing material for heaters with fan and smoke jack assembly

FIND NO.	IDENTIFICATION	DESCRIPTION	LEVEL A	LEVEL B	LEVEL C
1	PPP-B-601, overseas type, style A, type 3 load (see Note 1)	Box, Plywood, Cleated	1		
1	PPP-B-591, class II, style A (see note 1)	Box, Fiberboard, Cleated		1	1
2	PPP-B-636, V3c, style RSC	Box, Fiberboard	1	1	
2	PPP-B-636, type CF, class domestic, variety SW, grade 275, style RSC	Box, Fiberboard			1
3	PPP-F-320, type CF, class domestic, variety SW, grade 125	Fiberboard	AR	AR	AR
4	Base, side members	Lumber, softwood, 1x3 nominal	2	2	2
5	Base, front cross member	Lumber, softwood, 1x6 nominal	1	1	1
6	Base, rear cross member	Lumber, softwood, 1x3 nominal	1	1	1
7	FF-N-105, type II, style 4, table Vb	Nail, 6d	AR	AR	AR
8	MS27183-11, .312 IDx.734 ODx.065 THK	Washer, flat-round, steel, cad plated, general purpose	4	4	4
9	MS35338-45, 5/16 nominal size	Washer, lockspring, helical, regular series, stl, cad plated	4	4	4
10	MS35206-301, 5/16-18UNC-2Ax1-3/4 long	Screw, mach, pan hd, cross recessed, cs, cad plated	4	4	4
11	PPP-T-60, type III, class I	Tape, pressure sensitive	AR	AR	AR
12	QQ-W-461, finish 5, soft, class 1, .080 DIA	Wire, light, soft tensile, light zinc coated	AR	AR	AR
13	PPP-C-843, type II, class A	Cushioning	AR	AR	AR
14	Blocking	Lumber, softwood, 1x3 nominal	3	3	3
15	QQ-S-781, Class I, type I or IV, Finish A	Steel strapping, flat, .023 x 1/2 W	AR	AR	AR

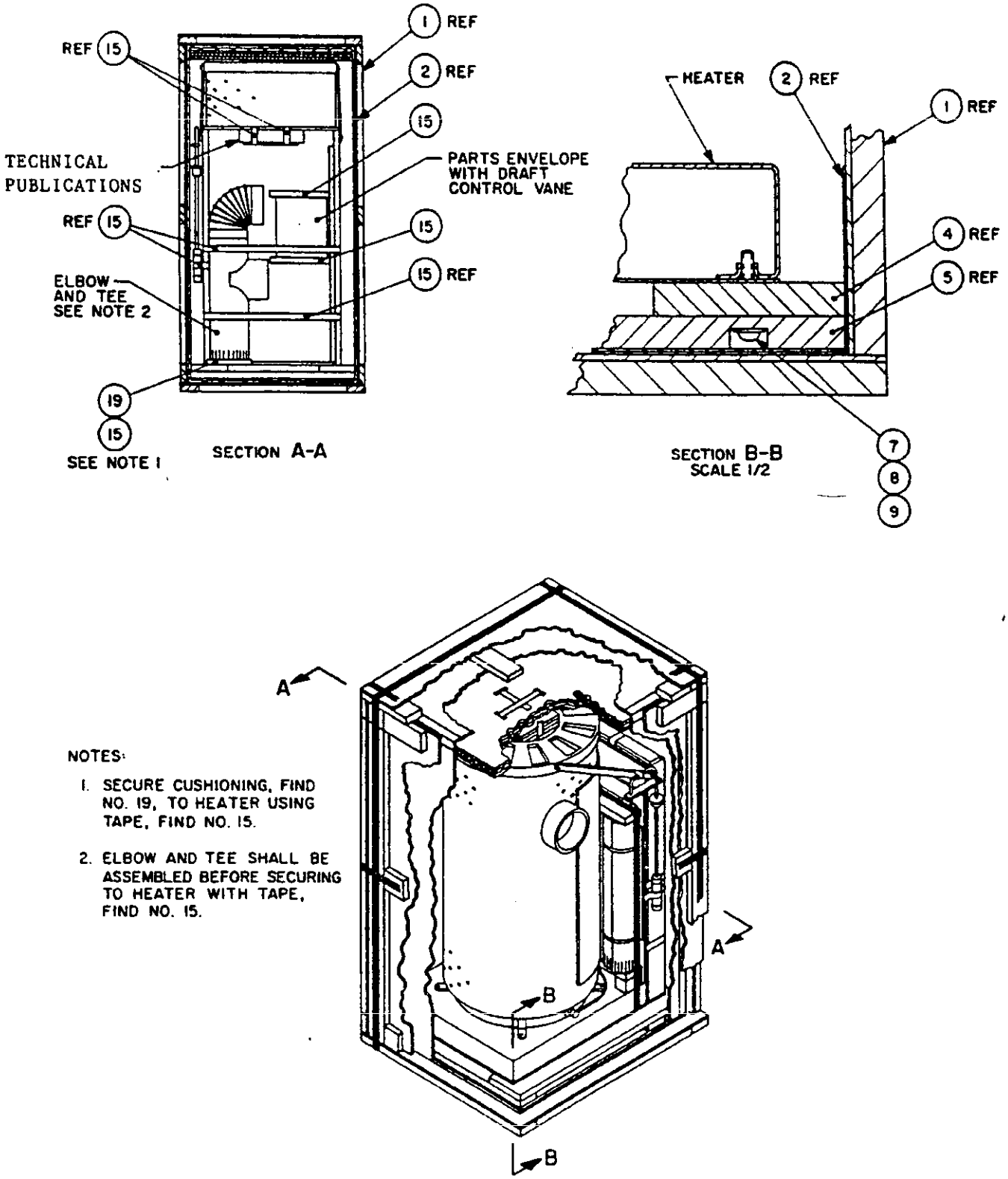
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TABLE V. Preservation and packing material for heaters with fan and smoke jack assembly (cont'd)

FIND NO.	IDENTIFICATION	DESCRIPTION	LEVEL A	LEVEL B	LEVEL C
16	PPP-B-636, W5c	Box, Fiberboard	1		
16	PPP-B-636, type CF, class domestic variety SW, grade 175	Box, Fiberboard		1	1
17	PPP-F-320, Type CF, class domestic, variety TW, grade 1100	Fiberboard	AR	AR	AR
18	PPP-T-97, type III, 3/4 w	Tape, pressure sensitive,	AR	AR	AR
19	QQ-S-781, class I, type I or IV, Finish A	Steel strapping, flat, .035 x 3/4 W	AR	AR	AR
20	MS24629-48, number 10-24 x 3/4 LG	Screw, tapping-THD cutting, type D, pan hd, cross recessed, CS, CAD, PLD	2	2	2
21	Blocking	Lumber, softwood, 2x2 nominal	AR	AR	AR
22	PPP-B-636, W5c	Box, Fiberboard	1		
22	PPP-B-636, type CF, class domestic, variety SW, grade 175	Box, Fiberboard		1	1

NOTES:
1. Top to be inverted.
2. AR denotes "as required".

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NOTES:

1. SECURE CUSHIONING, FIND NO. 19, TO HEATER USING TAPE, FIND NO. 15.
2. ELBOW AND TEE SHALL BE ASSEMBLED BEFORE SECURING TO HEATER WITH TAPE, FIND NO. 15.

FIGURE 1. Packing arrangement and details for heaters without fan and smoke jack assembly.

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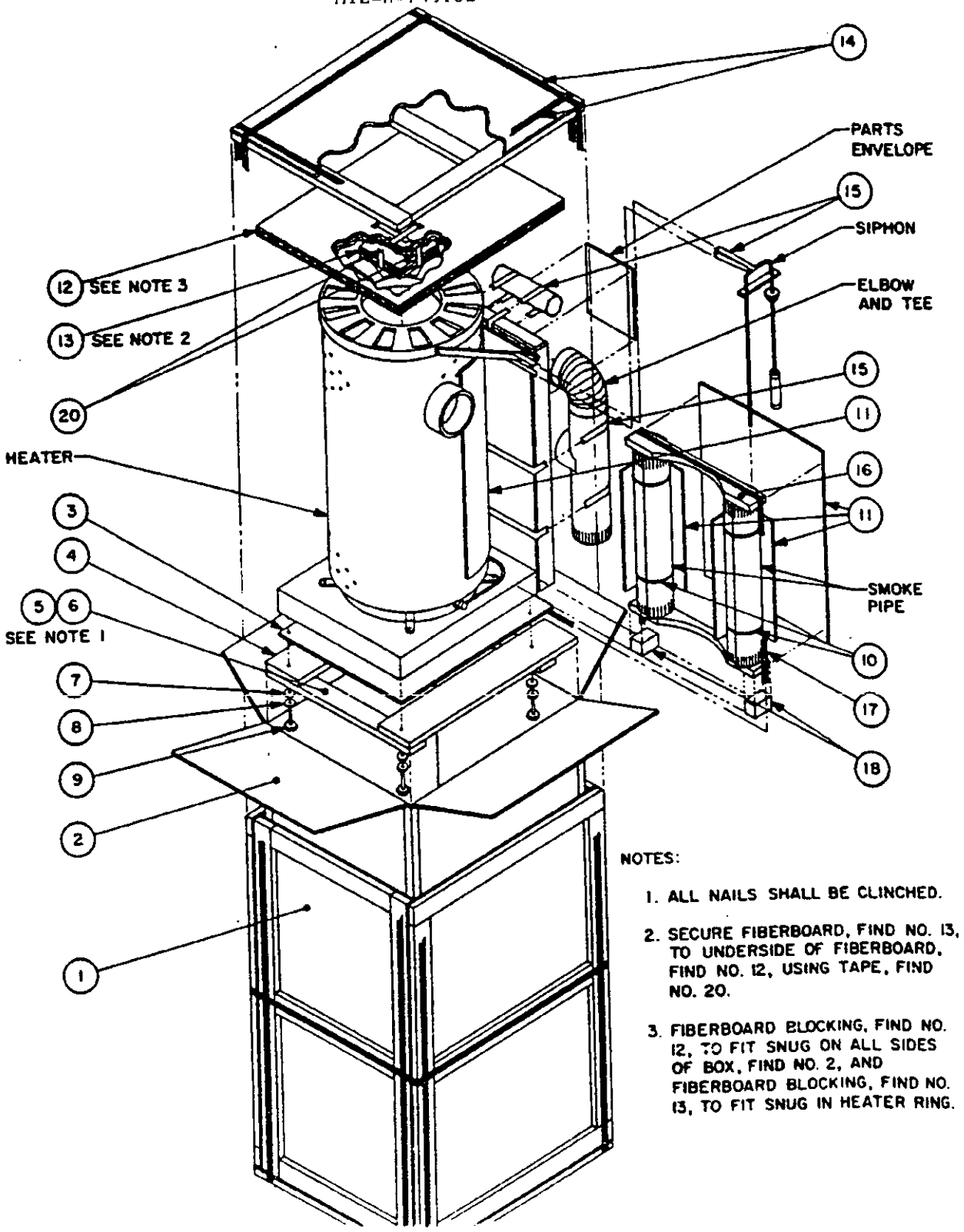


FIGURE 2. Packing assembly for heaters without fan and smoke jack assembly.

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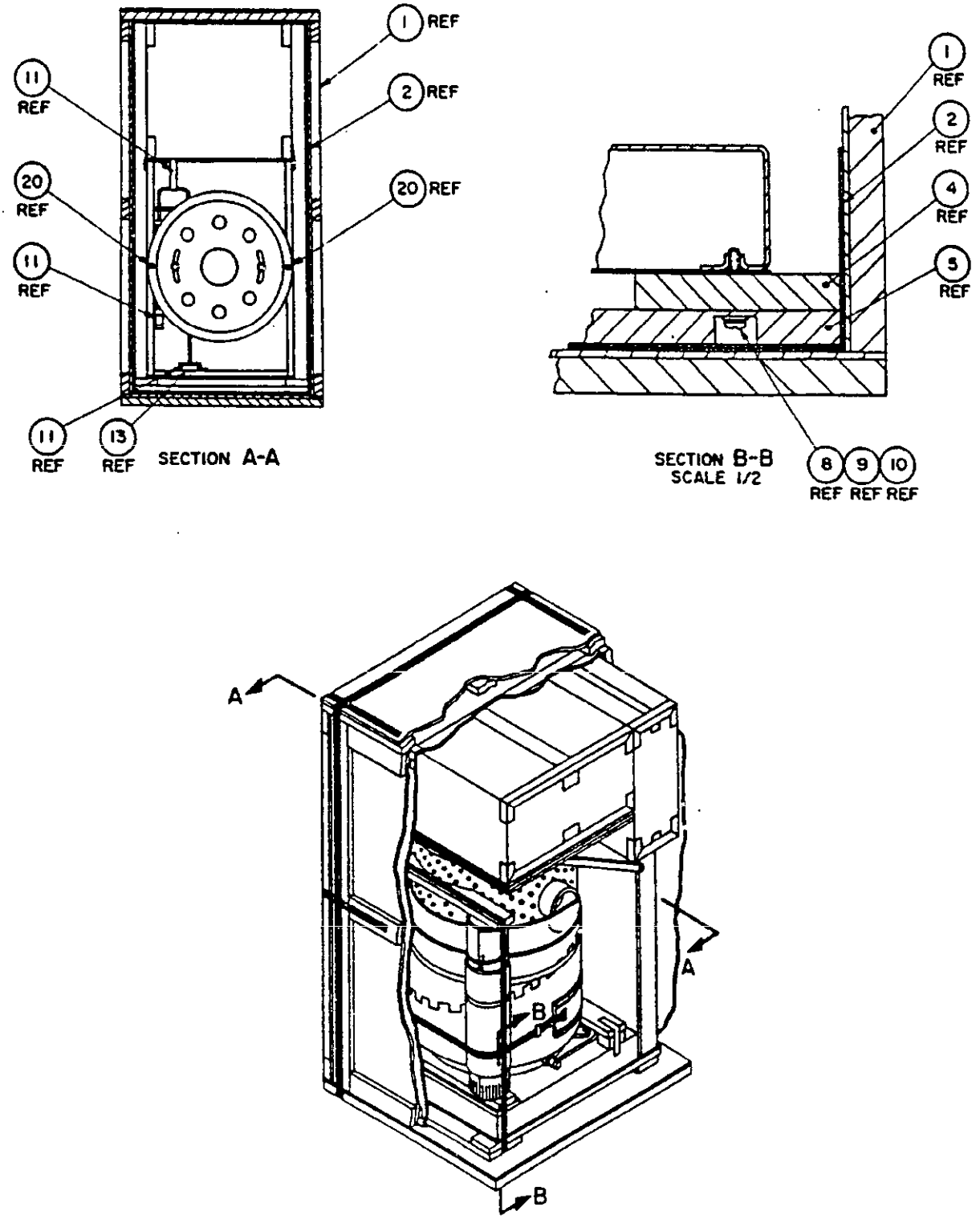


FIGURE 3. Packing arrangement and details for heaters with fan and smoke jack assembly.

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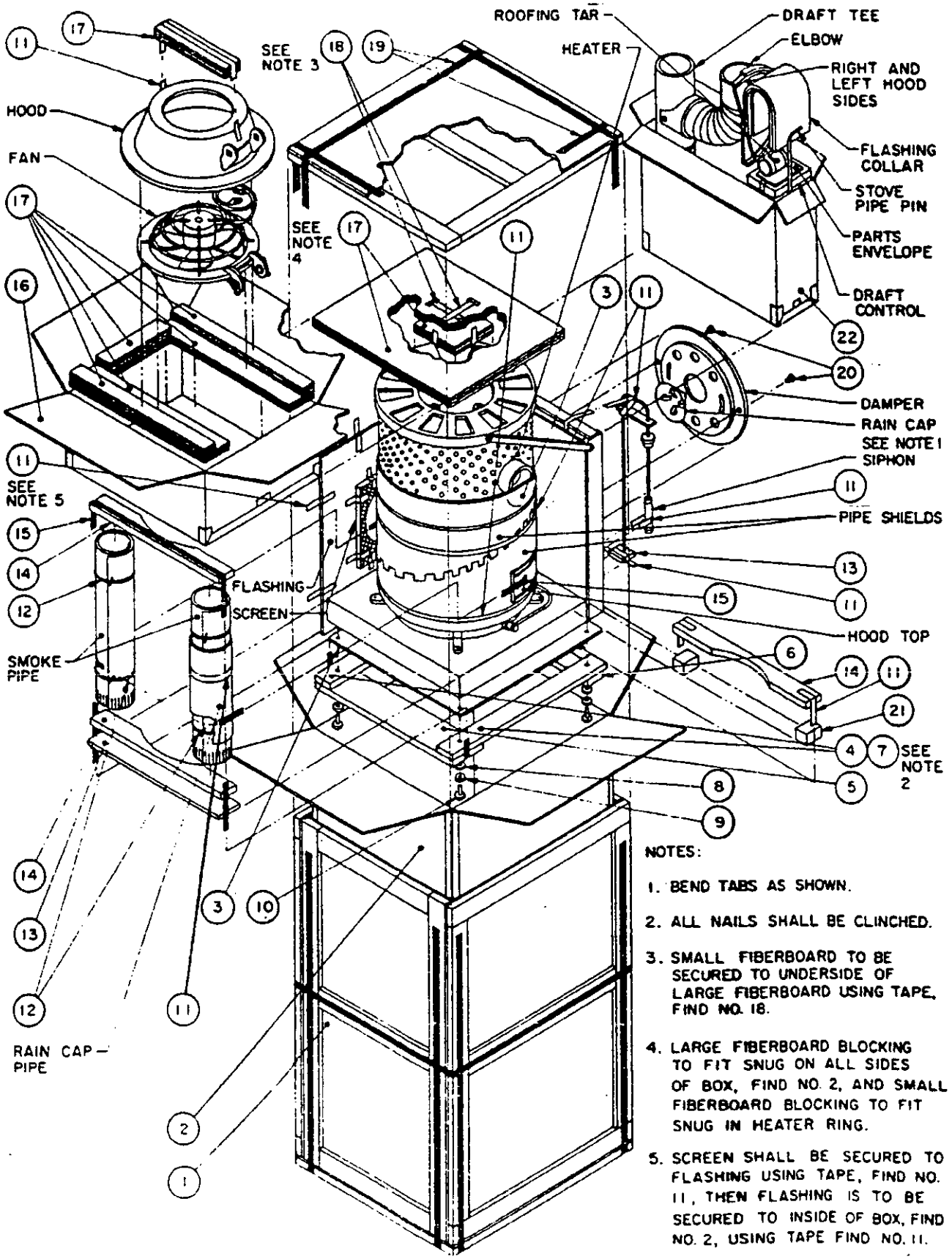


FIGURE 4. Packing assembly for heaters with fan and smoke jack assembly.

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(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-H-14516E	Heater, Space: Oil-Fired 70,000 BTU Per Hour, Theater of Operations Type

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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